# Failure Analysis Report

FAIL-0559 — Synchronization Failure

## 1. Failure Summary

| | |
|---|---|
| **Failure ID** | FAIL-0559 |
| **Failure Type** | Synchronization Failure |
| **Severity** | Critical |
| **Component** | Voice Pipeline |
| **Application** | APP-003 — GalaxySearch |
| **Build Version** | 4.4.5 |
| **Timestamp** | 2024-07-01 18:40:37 |
| **Environment** | ENV-007 — Staging |
| **OS Version** | Android 13 |
| **Network** | LTE |
| **Resolution** | Wont Fix |

## 2. Test Case Reference

| | |
|---|---|
| **Test Case ID** | TC-0697 |
| **Title** | Ensure that configuration reload does not exceed the p99 latency (1s) threshold ... |
| **Execution ID** | EXEC-02872 |
| **Test Run** | RUN-005 — Regression Cycle 5 — Build 5.1.6 |
| **Requirement** | REQ-0309 — Compatibility — Firmware Over-The-Air Update |

## 3. Expected vs Observed Behaviour

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|---|---|
| **Expected Behaviour** | The system should complete session token refresh within the defined SLA. |
| **Observed Behaviour** | The system exhibited synchronization failure when two concurrent requests are made simultaneously. |

## 4. Error Details

Error Message:

```
SyncError: conflict detected on entity 'ENTITY-6034'; last_modified timestamps diverge by 13s.
```

Stack Trace / Log Excerpt:

```
com.samsung.wallet.NetworkStack.execute(line 763)
```

| | |
|---|---|
| **Log Reference** | LOG-72372 |

## 5. Executed Steps at Time of Failure

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| | | |
|---|---|---|
| 1 | Observe the status indicator for up to 28 seconds. | Server returns HTTP 200 with a valid JSON payload. |
| 2 | Check server logs for event code 'EVT-5897'. | The performance counter logs a value below the SLA threshold. |
| 3 | Enter valid credentials: username='testuser_014', password='[REDA... | The device reconnects automatically within 10 seconds. |
| 4 | Re-connect the wearable and confirm auto-recovery. | A confirmation notification appears on the paired device. |

## 6. Root Cause Analysis

UI thread blocked by synchronous network call; ANR triggered after 5 seconds.

## 7. Linked Defect

| | |
|---|---|
| **Defect ID** | BUG-0115 |
| **Title** | Synchronization Failure in Voice Pipeline: SyncError: conflict detected on entit... |
| **Severity** | Critical |
| **Priority** | Critical |
| **Status** | REOPENED |
| **Root Cause** | UI thread blocked by synchronous network call; ANR triggered after 5 seconds. |
| **Resolution** | — |
| **Assigned To** | ENG-043 |

## 8. Historical Similar Failures

| | | | | | | |
|---|---|---|---|---|---|---|
| HIST-0076 | FAIL-0077 | Synchronization Failure | 6.2.0 | 2023-11-24 | 4 | Open |
| HIST-0095 | FAIL-0096 | Synchronization Failure | 6.5.4 | 2022-01-14 | 13 | Open |
| HIST-0407 | FAIL-0408 | Synchronization Failure | 4.3.0 | 2023-02-21 | 3 | Resolved |

## 9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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